

Assuming your existing files are:

```
/src
├─ CarbonEngine.js
├─ constants.js
├─ storageController.js
├─ app.js
├─ styles.css
└─ index.html
```

and that **CarbonEngine** exposes:

```
calculateDailyFootprint(input)
getPersonalizedInsight(input)
```

here is a complete lightweight implementation.

storageController.js

```
const STORAGE_KEY = "cft_state";
```

```
const DEFAULT_STATE = {
  lastUpdated: null,
  inputs: {
    transport: {
      car: 0,
      bus: 0,
      train: 0,
      bike: 0
    },
    meals: {
      meat: 0,
      vegetarian: 0,
      vegan: 0
    },
    energy: 0
  },
  results: {
    totalKgCO2e: 0,
    equivalents: {},
    insight: ""
  }
};
```

```
export const storageController = {
  get() {
    try {
      const stored = localStorage.getItem(STORAGE_KEY);

      if (!stored) {
        localStorage.setItem(
          STORAGE_KEY,
          JSON.stringify(DEFAULT_STATE)
        );
        return DEFAULT_STATE;
      }

      return JSON.parse(stored);
    } catch (error) {
      console.error("Storage read error:", error);
      return DEFAULT_STATE;
    }
  },

  save(data) {
    try {
      localStorage.setItem(
        STORAGE_KEY,
        JSON.stringify(data)
      );
    } catch (error) {
      console.error("Storage save error:", error);
    }
  }
};
```

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta
    name="viewport"
    content="width=device-width, initial-scale=1.0"
  />
  <title>Carbon Footprint Tracker</title>

  <link rel="stylesheet" href="/styles.css" />
```

```
</head>
<body>

<main class="container">

  <header class="hero">
    <h1>🌿 Carbon Footprint Tracker</h1>
    <p>Understand your daily environmental impact.</p>
  </header>

  <section class="card">

    <h2>Log Today's Activity</h2>

    <form id="footprint-form">

      <div class="section-group">
        <h3>Transportation (km)</h3>

        <label>
          Car
          <input type="number" min="0" value="0" id="carKm" />
        </label>

        <label>
          Bus
          <input type="number" min="0" value="0" id="busKm" />
        </label>

        <label>
          Train
          <input type="number" min="0" value="0" id="trainKm" />
        </label>

        <label>
          Bike
          <input type="number" min="0" value="0" id="bikeKm" />
        </label>
      </div>

      <div class="section-group">
        <h3>Meals</h3>

        <label>
          Meat Meals
          <input type="number" min="0" value="0" id="meatMeals" />
        </label>
```

```
<label>
  Vegetarian Meals
  <input type="number" min="0" value="0" id="vegetarianMeals" />
</label>
```

```
<label>
  Vegan Meals
  <input type="number" min="0" value="0" id="veganMeals" />
</label>
</div>
```

```
<div class="section-group">
  <h3>Energy Usage</h3>
```

```
  <label>
    kWh Consumed
    <input
      type="number"
      min="0"
      step="0.1"
      value="0"
      id="energyUsed"
    />
  </label>
</div>
```

```
<button type="submit" class="btn-primary">
  Calculate Footprint
</button>
```

```
</form>
</section>
```

```
<section class="dashboard">
```

```
<div class="dashboard-card">
  <h2>Daily Dashboard</h2>
```

```
  <div class="footprint-number">
    <span id="totalFootprint">0.00</span>
    <small>kg CO2e</small>
  </div>
```

```
  <div id="equivalents" class="equivalents">
    Start logging activities to see your impact.
  </div>
</div>
```

```
<div class="insight-card">
  <h2>Your Daily Mix</h2>

  <p id="dailyInsight">
    Your personalized sustainability insight will appear here.
  </p>
</div>

</section>

</main>

<script type="module" src="./app.js"></script>

</body>
</html>
```

styles.css

```
:root {
  --green-dark: #2f6f57;
  --green-main: #4f9b78;
  --green-light: #dff3e7;
  --bg: #f7faf8;
  --card: #ffffff;
  --text: #27323f;
  --border: #dce7df;
  --shadow: 0 10px 25px rgba(0,0,0,0.06);
}

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  background: var(--bg);
  color: var(--text);
  font-family:
    Inter,
    system-ui,
    sans-serif;
  line-height: 1.5;
}
```

```
.container {  
  width: min(1100px, 92%);  
  margin: 40px auto;  
}
```

```
.hero {  
  text-align: center;  
  margin-bottom: 2rem;  
}
```

```
.hero h1 {  
  font-size: 2rem;  
  margin-bottom: 0.5rem;  
}
```

```
.card,  
.dashboard-card,  
.insight-card {  
  background: var(--card);  
  border-radius: 18px;  
  padding: 1.5rem;  
  box-shadow: var(--shadow);  
}
```

```
.card {  
  margin-bottom: 2rem;  
}
```

```
.section-group {  
  margin-bottom: 1.5rem;  
}
```

```
.section-group h3 {  
  margin-bottom: 0.75rem;  
  color: var(--green-dark);  
}
```

```
label {  
  display: flex;  
  flex-direction: column;  
  gap: 0.4rem;  
  margin-bottom: 0.9rem;  
  font-size: 0.95rem;  
}
```

```
input {  
  border: 1px solid var(--border);
```

```
border-radius: 10px;
padding: 0.8rem;
font-size: 1rem;
}
```

```
input:focus {
  outline: 2px solid var(--green-main);
  border-color: transparent;
}
```

```
.btn-primary {
  width: 100%;
  border: none;
  cursor: pointer;
  padding: 1rem;
  border-radius: 12px;
  background: var(--green-main);
  color: white;
  font-weight: 600;
  transition: 0.2s ease;
}
```

```
.btn-primary:hover {
  transform: translateY(-2px);
}
```

```
.dashboard {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 1.5rem;
}
```

```
.footprint-number {
  margin-top: 1rem;
  margin-bottom: 1rem;
}
```

```
.footprint-number span {
  display: block;
  font-size: 3rem;
  font-weight: 700;
  color: var(--green-dark);
}
```

```
.footprint-number small {
  font-size: 1rem;
}
```

```
.equivalents {
  margin-top: 1rem;
  color: #586370;
}

.insight-card {
  background:
    linear-gradient(
      135deg,
      #2f6f57,
      #5db187
    );
  color: white;
  border: none;
}

.insight-card h2 {
  margin-bottom: 1rem;
}

.insight-card p {
  font-size: 1.05rem;
  line-height: 1.7;
}

@media (max-width: 768px) {

  .dashboard {
    grid-template-columns: 1fr;
  }

  .footprint-number span {
    font-size: 2.5rem;
  }
}
```

app.js

```
import CarbonEngine from "../CarbonEngine.js";
import { storageController } from "../storageController.js";

const engine = new CarbonEngine();

const form = document.getElementById("footprint-form");
```



```

const totalFootprintEl =
  document.getElementById("totalFootprint");

const equivalentsEl =
  document.getElementById("equivalents");

const insightEl =
  document.getElementById("dailyInsight");

function calculateEquivalents(kgCO2e) {
  const smartphoneCharges = Math.round(
    kgCO2e * 122
  );

  const ledBulbHours = Math.round(
    kgCO2e * 45
  );

  return {
    smartphoneCharges,
    ledBulbHours
  };
}

function updateDashboard(
  totalFootprint,
  equivalents,
  insight
) {
  totalFootprintEl.textContent =
    Number(totalFootprint).toFixed(2);

  equivalentsEl.innerHTML = `
    <p>📱 ≈ ${equivalents.smartphoneCharges} smartphone charges</p>
    <p>💡 ≈ ${equivalents.ledBulbHours} hours of LED lighting</p>
  `;

  insightEl.textContent = insight;
}

function loadSavedState() {
  const state = storageController.get();

  if (!state?.results) return;

  updateDashboard(
    state.results.totalKgCO2e || 0,
    state.results.equivalents || {

```

```
    smartphoneCharges: 0,  
    ledBulbHours: 0  
  },  
  state.results.insight || ""  
);  
}
```

```
form.addEventListener("submit", event => {  
  event.preventDefault();
```

```
  const inputData = {  
    transport: {  
      car: Number(  
        document.getElementById("carKm").value  
      ),  
      bus: Number(  
        document.getElementById("busKm").value  
      ),  
      train: Number(  
        document.getElementById("trainKm").value  
      ),  
      bike: Number(  
        document.getElementById("bikeKm").value  
      )  
    },
```

```
    meals: {  
      meat: Number(  
        document.getElementById("meatMeals").value  
      ),  
      vegetarian: Number(  
        document.getElementById("vegetarianMeals").value  
      ),  
      vegan: Number(  
        document.getElementById("veganMeals").value  
      )  
    },
```

```
    energy: Number(  
      document.getElementById("energyUsed").value  
    )  
  };
```

```
  const totalFootprint =  
    engine.calculateDailyFootprint(inputData);
```

```
  const insight =  
    engine.getPersonalizedInsight(inputData);
```

```

const equivalents =
  calculateEquivalents(totalFootprint);

const state = {
  lastUpdated: new Date().toISOString(),
  inputs: inputData,
  results: {
    totalKgCO2e: totalFootprint,
    equivalents,
    insight
  }
};

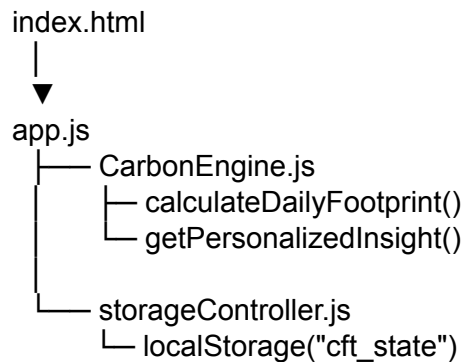
storageController.save(state);

updateDashboard(
  totalFootprint,
  equivalents,
  insight
);
});

loadSavedState();

```

Resulting architecture



`styles.css`

This gives you a fully client-side, zero-dependency carbon tracker with persistent local storage, responsive layout, dashboard metrics, and a highlighted behavioral-insight card.